ETHOS-VISION

The Autonomous Supply Chain Oracle

Empowering conscious consumption through real-time, autonomous ethical verification.

GLOBAL SUPPLY CHAINS ARE OPAQUE

70%

Consumers want to buy ethically

90%

Distrust corporate sustainability claims

- **Static Data:** Current "green scores" rely on outdated, self-reported, or incomplete corporate disclosures.
- **The Transparency Gap:** No real-time way to verify the ethical journey of a product at the point of purchase.
- **Information Asymmetry:** Corporations hold the data; consumers are left with marketing buzzwords.

ETHOS-VISION: THE "OUT OF THE BOX" SOLUTION

1. SCAN

Users scan a barcode or receipt to trigger an autonomous research swarm.

2. INVESTIGATE

A swarm of specialized AI agents autonomously crawls the web and verifies claims.

3. REVEAL

Complex data is transformed into a compelling, interactive ethical narrative.

"Beyond RAG: It doesn't just retrieve data; it autonomously investigates and verifies claims in real-time."

MULTI-MODAL VISION INTELLIGENCE

STATE-OF-THE-ART VISION

Leveraging Gemini 1.5 Pro and GPT-4o Vision for high-accuracy OCR and object recognition.

UNIVERSAL INPUT

Seamlessly processes barcodes, product labels, and store receipts to identify items.

FRICTIONLESS UX

Zero manual data entry. One snap triggers the entire autonomous research swarm.

"Turning a simple image into a gateway for global supply chain transparency."

CONTEXTUAL EXTRACTION

BRAND IDENTIFICATION

Parent Company & Subsidiary Mapping

PRODUCT LINE

Specific SKU & Manufacturing Batch

GEOGRAPHIC ORIGIN

Sourcing Region & Logistics Hubs

THE AUTONOMOUS RESEARCH SWARM

AGENTIC ORCHESTRATION POWERED BY LANGGRAPH & CREWAI

LABOR AUDITOR

Investigates factory conditions, wage reports, and labor union disclosures to identify human rights risks.

CARBON TRACKER

Estimates logistics and manufacturing emissions by cross-referencing shipping routes and energy grid data.

NEWS FACT-CHECKER

Scans global news archives and NGO databases for recent ethical controversies or environmental violations.

TRUST VERIFIER

Validates corporate claims against independent third-party certifications and satellite imagery evidence.

REAL-TIME VERIFICATION & FACT-CHECKING

DYNAMIC RESEARCH SOURCES

- Live Web Crawl
- Financial Reports
- Satellite Imagery
- News Archives
- NGO Databases
- Legal Filings

DYNAMIC ETHOS SCORE: 85/100

Verified against 14 independent sources

VERIFICATION LOGIC

Cross-Referencing: Agents validate corporate marketing claims against independent news and financial data.

Conflict Detection: Identifies discrepancies between self-reported sustainability and actual supply chain behavior.

Evidence-Based: Every score component is backed by a specific, verifiable source link for the user to explore.

"Moving from blind trust to verified transparency through autonomous agentic reasoning."

FROM DATA TO INTERACTIVE NARRATIVE

PERSONALIZED STORYTELLING

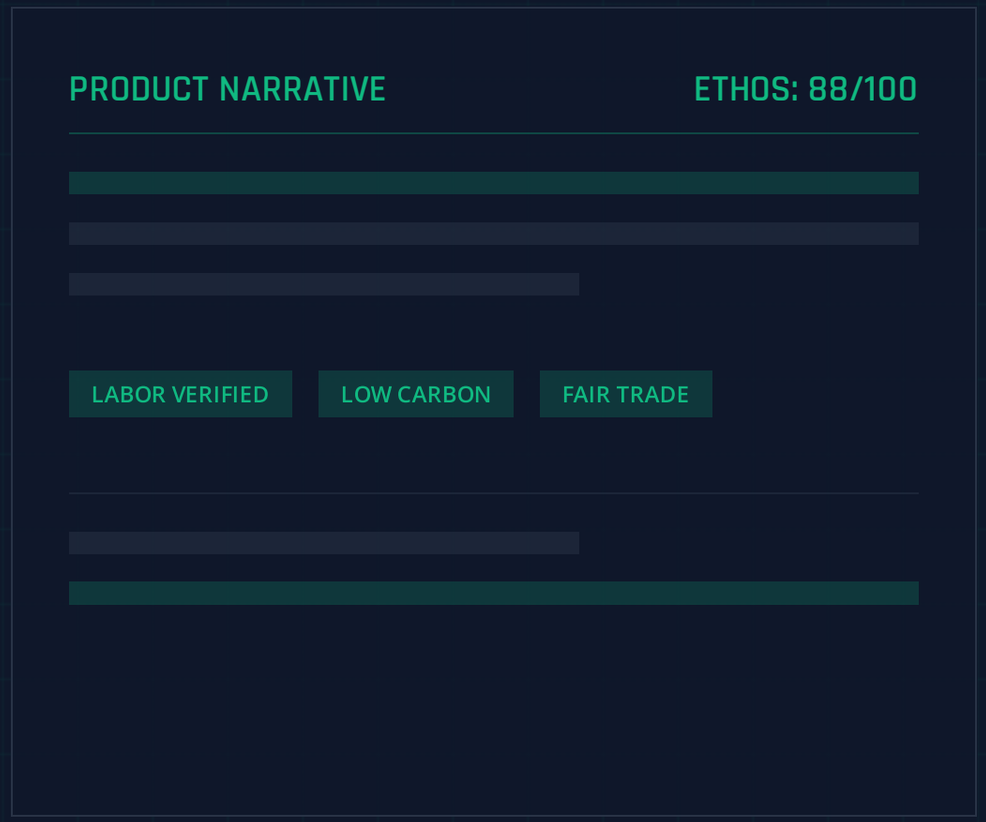
Converts complex supply chain data into a compelling, easy-to-understand narrative tailored to user values.

DRILL-DOWN CAPABILITIES

Users can interact with specific agent findings to view source documents, news reports, or carbon calculations.

ACTIONABLE INSIGHTS

Suggests ethical alternatives in real-time if the scanned product fails the user's personal criteria.



TECHNICAL ARCHITECTURE & SCALABILITY

FRONTEND

- **React / Tailwind:** Sleek, responsive web application.
- **Scan-to-Story:** Intuitive mobile-first user experience.
- **Real-time UI:** Live progress bars for agent research.

BACKEND

- **Python Framework:** High-performance agentic core.
- **LangGraph:** Complex multi-agent orchestration.
- **Vision LLMs:** Gemini/GPT-4o for multi-modal OCR.

SCALABILITY

- **Intelligent Caching:** Storing verified findings for reuse.
- **Periodic Updates:** Agents refresh data asynchronously.
- **Distributed Swarms:** Horizontal scaling of agent tasks.

DESIGNED FOR MILLIONS OF PRODUCTS & REAL-TIME VERIFICATION